MTH201 - Multivariable Calculus (Quiz No.4)

Question # 1 of 10 (Start time: 02:03:56 PM, 04 July 2025)

In general, the value of the line integral depends on the particular path of integration.

Select the correct option

○	False
◉	True

MTH201 - Multivariable Calculus (Quiz No.4)

Question # 4 of 10 (Start time: 02:04:39 PM, 04 July 2025)

The value of the integral is --------- independent of the path of integration taken.

Select the correct option

○	always
◉	may or may not

MTH201 - Multivariable Calculus (Quiz No.4)

Question # 9 of 10 (Start time: 02:06:56 PM, 04 July 2025)

Evaluate $\int lnxdx = - - - - - - - - -$

Select the correct option

○	$lnx - \frac{1}{x} + c$
○	$lnx - x + x$
●	$xlnx - x + c$
○	$xlnx - \frac{1}{x} + c$

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Question # 7 of 10 (Start time: 02:06:05 PM, 04 July 2025)

The path traversal in calculating the Green's Theorem is --------

Select the correct option

○	outwards
●	anticlockwise
○	clockwise
○	inwards

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Question # 7 of 10 (Start time: 02:00:05 PM, 04 July 2025)

Integration along two separate paths joining the same two end points does not necessarily give identical results.

Select the correct option

◉	True
○	False

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Question # 5 of 10 (Start time: 02:04:51 PM, 04 July 2025)

If $I = \int_{AB} Pdx + Qdy$ and $(Pdx + Qdy)$ is an exact differentialthen

Select the correct option

◉	$I_{c_1} = I_{c_2}$
○	$I_{c_2} = I_{c_2}$
○	$I_{c_1} = 2I_{c_1}$
○	$I_{c_1} = -I_{c_2}$

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Question # 5 of 10 (Start time: 01:59:24 PM, 04 July 2025)

Wallis sine formula when n is even

$$\int_0^{\frac{\pi}{2}} Sin^4 x dx =$$

Select the correct option

○	$\frac{4}{5} \cdot \frac{2}{3}$
○	$\frac{3}{4} \cdot \frac{1}{2}$
○	$\frac{4}{3} \cdot \frac{2}{1} \cdot \frac{\pi}{2}$
◉	$\frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2}$

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Question # 9 of 10 (Start time: 02:00:31 PM, 04 July 2025)

If $I = \int_{AB} Pdx + Qdy$ and $(Pdx + Qdy)$ is an exact differentialthen

Select the correct option

○	$I_{c_1} = -I_{c_2}$
◉	$I_{c_1} = I_{c_2}$
○	$I_{c_2} = I_{c_2}$
○	$I_{c_1} = 2I_{c_1}$

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Question # 10 of 10 (Start time: 02:00:52 PM, 04 July 2025)

$$\int_0^{\frac{\pi}{2}} Cos^2 x dx = \frac{1}{2}\left|\frac{\pi}{2} + \frac{\sin \pi}{2}\right| =$$

Select the correct option

○	$\frac{3\pi}{4}$
●	$\frac{\pi}{4}$
○	$\frac{\pi}{2}$
○	$\frac{\pi}{3}$

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Question # 3 of 10 (Start time: 02:04:24 PM, 04 July 2025)

$$\text{If} I = \int_{AB} Pdx + Qdy \text{ and } (Pdx + Qdy) \text{ is an exact differentialthen}$$

Select the correct option

●	$I_{c_1} - I_{c_2} = 0$
○	$I_{c_1} + I_{c_2} = 0$
○	$I_{c_1} \times I_{c_2} = 0$
○	$I_{c_2} + I_{c_2} = 0$

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Question # 8 of 10 (Start time: 02:00:18 PM, 04 July 2025)

Path of integration parallel to _____, $dx = 0$. $\therefore\ I_C = \int_C Q\,dy.$

Select the correct option

○	z - axis
○	x - axis
○	$dz = 0$
◉	y - axis

MTH201 - Multivariable Calculus (Quiz No.4)

Question # 10 of 10 (Start time: 02:07:16 PM, 04 July 2025)

The integration taken round a closed curve is ------- provided $(Pdx + Qdy)$ is a(n) ---------- differential.

Select the correct option

○	zero, homogeneous
○	one, exact
◉	zero, exact
○	one, homogeneous

MTH201 - Multivariable Calculus (Quiz No.4)

Question # 1 of 10 (Start time: 01:58:00 PM, 04 July 2025)

If$(Pdx + Qdy)$ is an exact differentialthen

Select the correct option

- $\oint (Pdx + Qdy) = 1$
- $\oint (Pdx + Qdy) = -1$
- $\oint (Pdx + Qdy) = 0$
- $\oint (Pdx + Qdy) \neq 0$

MTH201 - Multivariable Calculus (Quiz No.4)

Question # 2 of 10 (Start time: 02:04:09 PM, 04 July 2025)

To evaluate the line integral, $\int_C V(r)\,dr$, the integrand is expressed in terms of x, y, z with $d\vec{r} =$ ____

Select the correct option

- $dx\,\hat{i} + dy\,\hat{j}$
- $\sqrt{dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k}}$
- $dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k}$
- $dx\,\hat{i}$

Click to

MTH201 - Multivariable Calculus (Quiz No.4)

Question # 6 of 10 (Start time: 01:59:40 PM, 04 July 2025)

Wallis sine formula when n is odd

$$\int_0^{\frac{\pi}{2}} Cos^7 x dx =$$

Select the correct option

○	$\frac{7}{6} \cdot \frac{5}{4} \cdot \frac{3}{2} \cdot \frac{\pi}{2}$
○	$\frac{7}{6} \cdot \frac{5}{4} \cdot \frac{3}{2}$
◉	$\frac{6}{7} \cdot \frac{4}{5} \cdot \frac{2}{3}$
○	$\frac{6}{7} \cdot \frac{4}{5} \cdot \frac{2}{3} \cdot \frac{\pi}{2}$

MTH201 - Multivariable Calculus (Quiz No.4)

Question # 4 of 10 (Start time: 01:59:05 PM, 04 July 2025)

To eveluate a line integral, the integrand is expressed in terms of x, y, z with

Select the correct option

○	$dr = x + y + z$
○	$dr = dx + dy + dz$
◉	$dr = dxi + dyj + dzk$
○	$dr = xi + yj + zk$

Question # 6 of 10 (Start time: 02:05:48 PM, 04 July 2025)

Consider a vector A for which $\nabla . A = 0$ at all points, i,e. for all values of x, y, z, is called a(an) _____.

Select the correct option

◉	solenoid vector
○	scalar
○	unit vector
○	constant

Question # 3 of 10 (Start time: 01:58:54 PM, 04 July 2025)

If the integrand of the given integral is seen to be a (an) _____ differential, then the value of the line integral is independent of the path taken.

Select the correct option

◉	exact
○	non-exact
○	partial
○	total

MTH201 - Multivariable Calculus (Quiz No.4)

Question # 8 of 10 (Start time: 02:06:16 PM, 04 July 2025)

Wallis sine formula when n is even

$$\int_0^{\frac{\pi}{2}} Sin^n x dx =$$

Select the correct option

○	$\frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdot \frac{n-5}{n-4} \cdot \frac{n-7}{n-6} \cdots \frac{6}{7} \cdot \frac{4}{5} \cdot \frac{2}{3}$
○	$\frac{n}{2} \cdot \frac{n-2}{2} \cdot \frac{n-4}{2} \cdot \frac{n-6}{2} \cdots \frac{6}{7} \cdot \frac{4}{5} \cdot \frac{2}{3}$
◉	$\frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdot \frac{n-5}{n-4} \cdot \frac{n-7}{n-6} \cdots \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2}$
○	$\frac{n-1}{2} \cdot \frac{n-1}{2} \cdot \frac{n-1}{2} \cdot \frac{n-1}{2} \cdots \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2}$

MTH201 - Multivariable Calculus (Quiz No.4)

Question # 2 of 10 (Start time: 01:58:17 PM, 04 July 2025)

For line integral with respect to arc length, when x and y are expressed in parametric form,

$$I = \int_C f(x,y)ds = \int_{t_1}^{t_2} f(x,y)ds, \text{ where } ds = _____.$$

Select the correct option

◉	$\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$
○	$\sqrt{\left(\frac{dx}{dt}\right)^2 - \left(\frac{dy}{dt}\right)^2}\, dt$
○	$\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$
○	$\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$

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